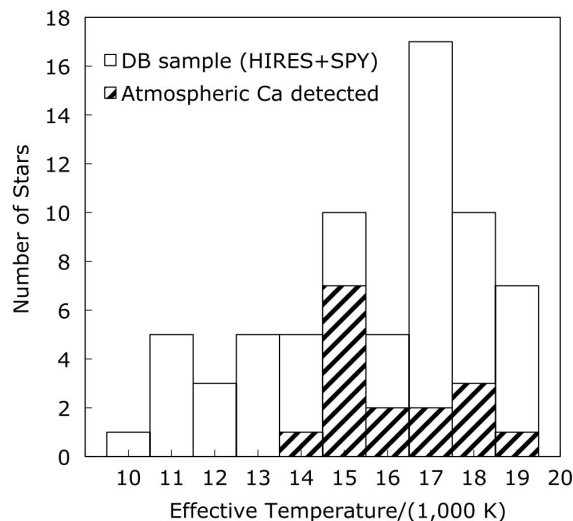


How to pollute a White Dwarf star - PHYS633

Matias Castro Tapia

Polluted White Dwarfs

- About 25% of white dwarfs show signatures of elements heavier than H and He in their atmospheres. Many of these with signatures of elements heavier than C (e.g., Eisenstein et al. 2006; Giammichele et al. 2012; Koester et al. 2014; Jiménez-Esteban et al. 2023)
- A graphic example from the sample presented in Zuckerman et al. (2010)



- Given the diffusion timescales of heavy elements these signatures can only be explained by recent accretion episodes onto the white dwarf atmosphere (e.g. Brouwers et al. 2022)
- The main explanations for pollutants are the accretion of planetary debris. However, other major objects can contribute to WD pollution (Veras 2020, for a review)

Planetary architectures around white dwarfs
Excluded are system-specific studies (Figs. 1-2)

- Simulations set-ups for studying white dwarf pollution from Veras (2020)
- In binaries the stability of up to two planets has been studied

# of stars	# of major planets	Main/Kuiper belt?	Oort comets?	Moons?	References
★					Veras, Eggli & Gänsicke (2015a) Makarov & Veras (2019)
★					Alcock, Fritsom & Siegelman (1986) Parriott & Alcock (1998) Veras, Shannon & Gänsicke (2014a) Stone, Metzger & Loeb (2015)
★					Grishin & Veras (2019)
★	P				Bonsor, Mustill & Wyatt (2011) Debes, Walsh & Stark (2012) Frewen & Hansen (2014) Antoniadou & Veras (2016, 2019)
★	P				Caiazzo & Heyl (2017)
★	P P				Debes & Sigurdsson (2002) Veras, Mustill, Bonsor et al. (2013) Voyatzis, Hadjidemetriou, Veras et al. (2013) Veras, Georgakarakos, Gänsicke et al. (2018a) Maldonado, Villaver, Mustill et al. (2020a)
★	P P P				Mustill, Veras & Villaver (2014) Maldonado, Villaver, Mustill et al. (2020b)
★	P P P				Mustill, Villaver, Veras et al. (2018) Smallwood, Martin, Livio et al. (2018)
★	P P P P +				Duncan & Lissauer (1998) Veras & Gänsicke (2015) Veras (2016b) Veras, Mustill, Gänsicke et al. (2016) Zink, Batygin & Adams (2020) Maldonado, Villaver, Mustill et al. (2021)
★	P P P P +				Veras, Reichert, Flammini Dotti et al. (2020b)
★	P P P P +				Payne, Veras, Holman et al. (2016) Payne, Veras, Gänsicke et al. (2017)
★ ★					Hamers & Portegies Zwart (2016)
★ ★	P				Kratter & Perets (2012) Veras & Tout (2012) Kostov, Moore, Tamayo et al. (2016) Stephan, Naoz & Gaudi (2018)
★ ★	P				Bonsor & Veras (2015) Petrovich & Muñoz (2017) Stephan, Naoz & Zuckerman (2017)
★ ★	P P				Portegies Zwart (2013) Mustill, Marshall, Villaver et al. (2013) Veras, Georgakarakos, Dobbs-Dixon et al. (2017b)

Fig. 6.— A rough breakdown of parameter space for investigations of planetary architectures around white dwarfs.

We observe systems that are polluted WD in wide binaries

A&A, 693, A64 (2025)

<https://doi.org/10.1051/0004-6361/202451621>

© The Authors 2025

**Astronomy
&
Astrophysics**

Host star and exoplanet composition: Polluted white dwarf reveals depletion of moderately refractory elements in planetary material★

Claudia Aguilera-Gómez^{1,2,★}, Laura K. Rogers³, Amy Bonsor³, Paula Jofré^{4,2}, Simon Blouin⁵,
Oliver Shorttle³, Andrew M. Buchan⁶, Yuqi Li³, and Siyi Xu⁷

¹ Instituto de Astrofísica, Pontificia Universidad Católica de Chile, Av. Vicuña Mackenna 4860, 782-0436 Macul, Santiago, Chile

² Millennium Nucleus ERIS, Chile

³ Institute of Astronomy, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK

⁴ Instituto de Estudios Astrofísicos, Universidad Diego Portales, Av. Ejército Libertador 441, Santiago, Chile

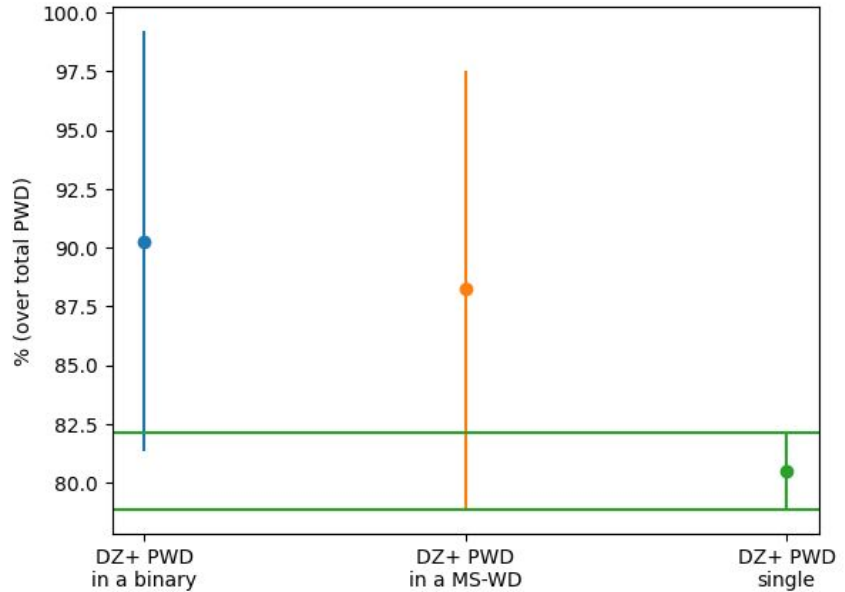
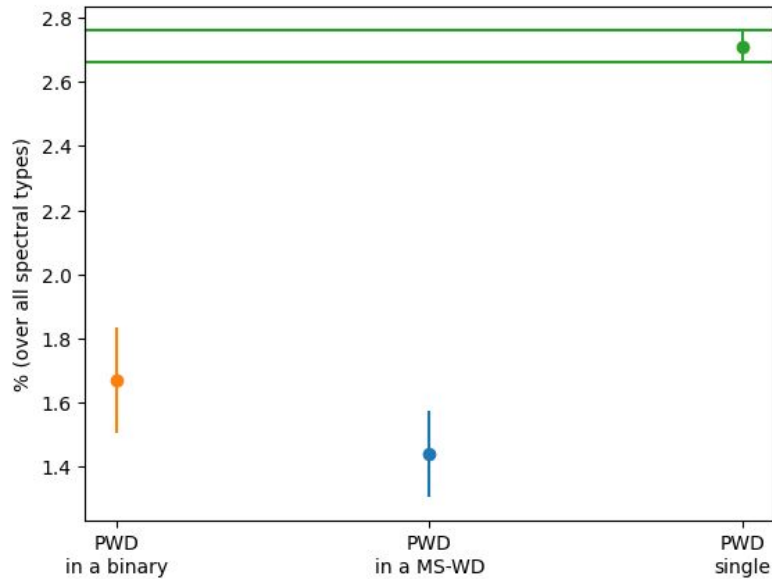
⁵ Department of Physics and Astronomy, University of Victoria, Victoria, BC V8W 2Y2, Canada

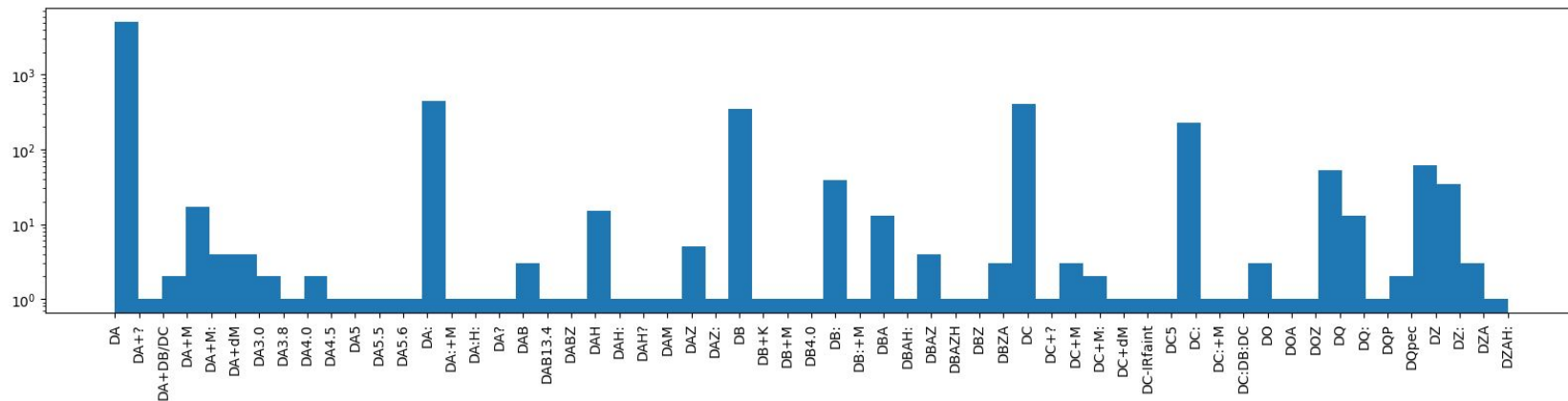
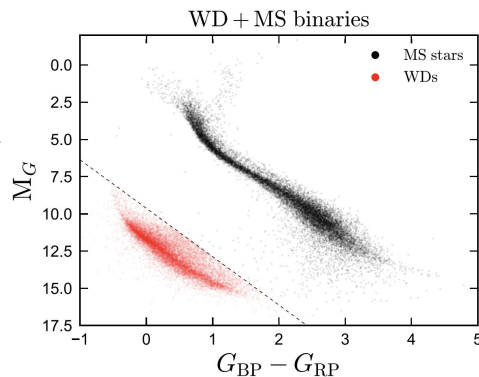
⁶ Department of Physics, University of Warwick, Coventry CV4 7AL, UK

⁷ Gemini Observatory/NSF's NOIRLab, 670 N. A'ohoku Place, Hilo, HI 96720, USA

Received 23 July 2024 / Accepted 1 November 2024

- Looking at matches in the wide binary catalog of El-Badry et al. (2021) and the Montreal White Dwarf Database (MWDD, Dufour et al, 2017)





Set-up

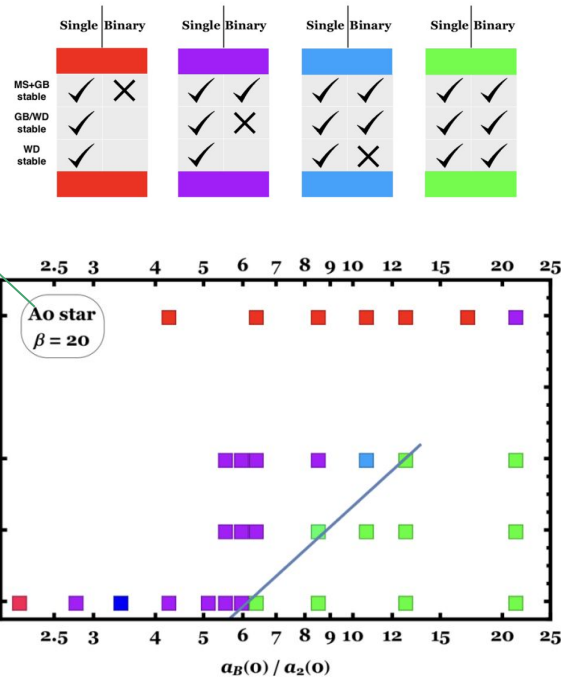
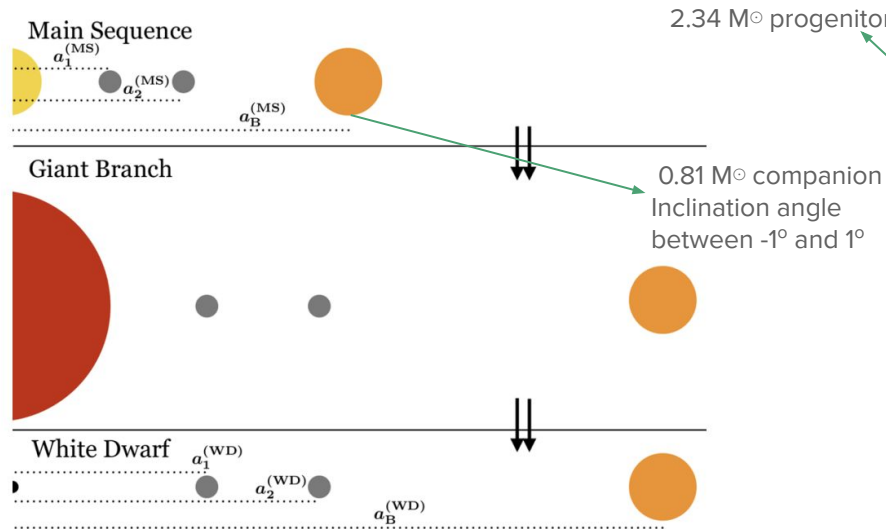
- Stability of planets for the whole stellar evolution has been studied for the case of four or more planets without a binary, and for only up to two planets in the case of a wide binary WD+MS

Binary star influence on post-main-sequence multi-planet stability

Dimitri Veras,¹★ Nikolaos Georgakarakos,² Ian Dobbs-Dixon² and Boris T. Gänsicke¹

¹Department of Physics, University of Warwick, Coventry CV4 7AL, UK

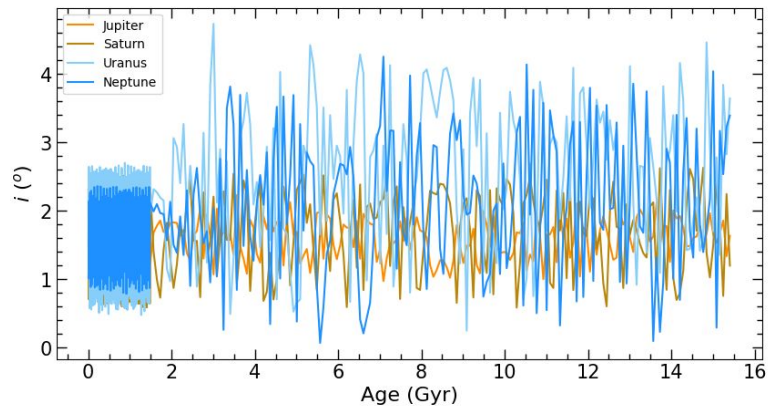
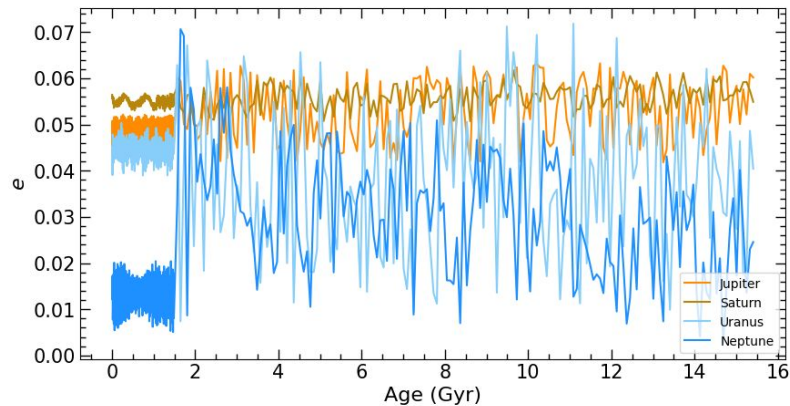
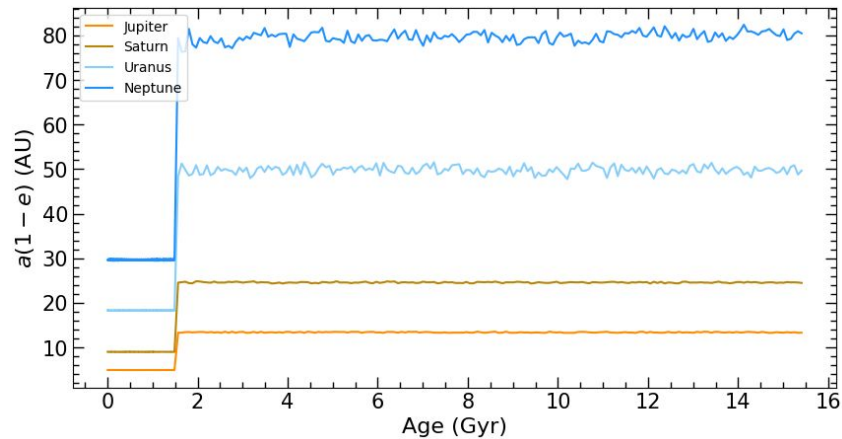
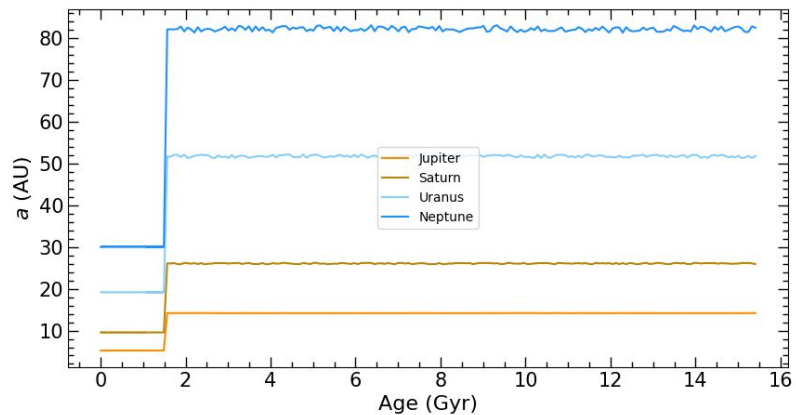
²New York University Abu Dhabi, Saadiyat Island, PO Box 129188, Abu Dhabi, UAE



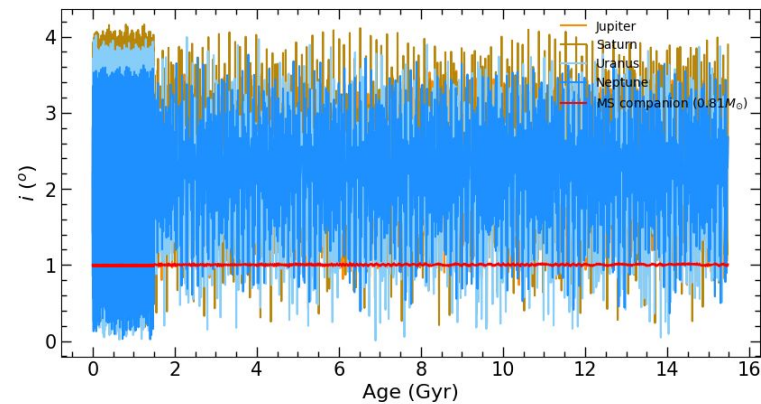
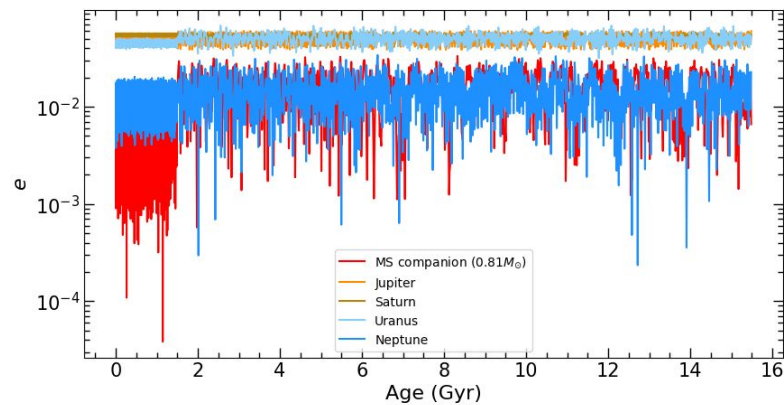
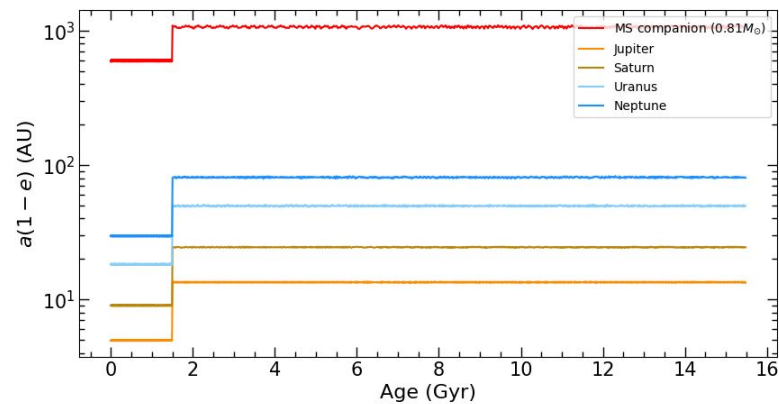
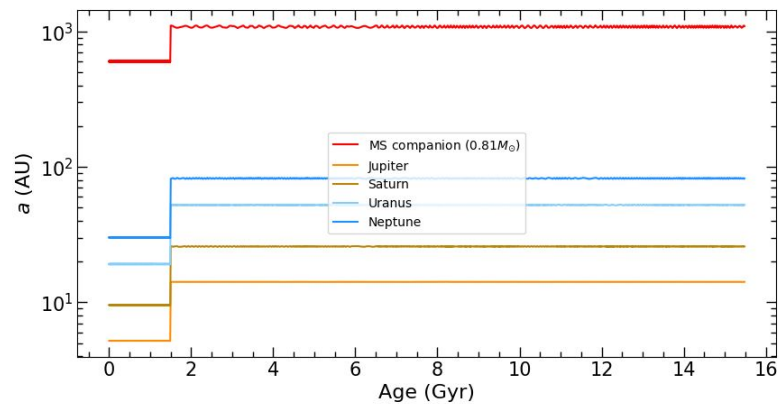
- I took two cases of a MS progenitor (for the WD) of $2M_{\odot}$ and $3M_{\odot}$ with a companion of $0.81M_{\odot}$ (evolved the systems for >14 Gyr)
- I took the orbital parameters of the four giant planets of the solar system, assuming that the Red Giant Phase of the progenitor will reach up to the Mars orbit (used Standish & Williams, 1992 formulation)
- I took $a_B/a_{\text{Neptune}}=20$, and I considered 4 different cases: no binary, inclinations of $i=1^{\circ}$, 20° , and 40°
- I skipped the giant phase considering that it occurs in a short timescale, but expanded the orbit of the system consistently with the mass loss of the star between the MS and the WD phase (Petrovich & Muñoz, 2016)

$$a_b(t) = a_b(0)(M_{s,0} + M_b)/(M_{s,0}e^{-t/\tau_{\text{ml}}} + M_b)$$

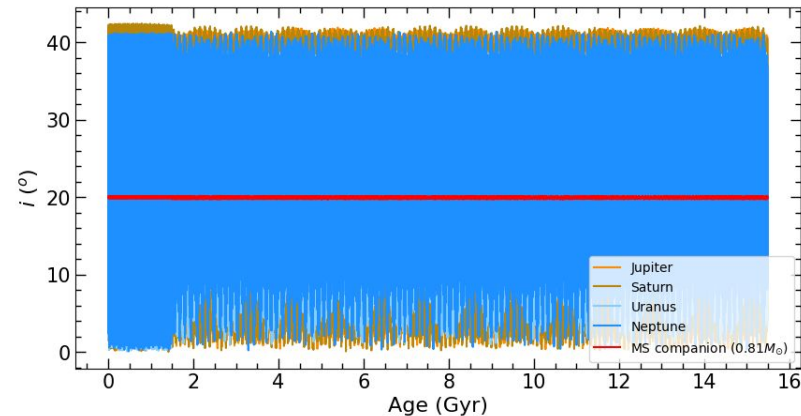
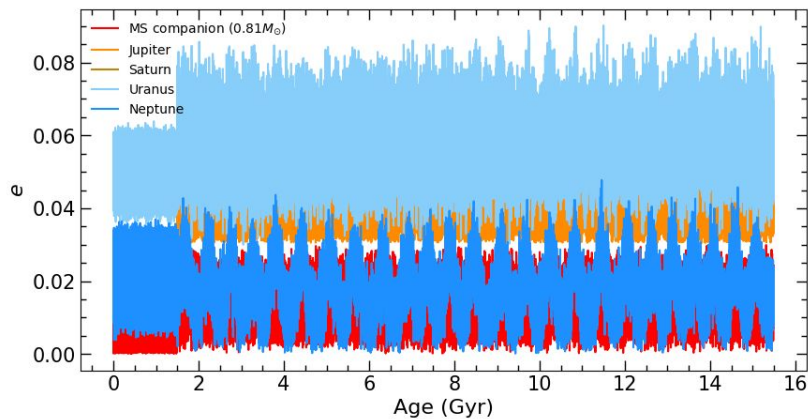
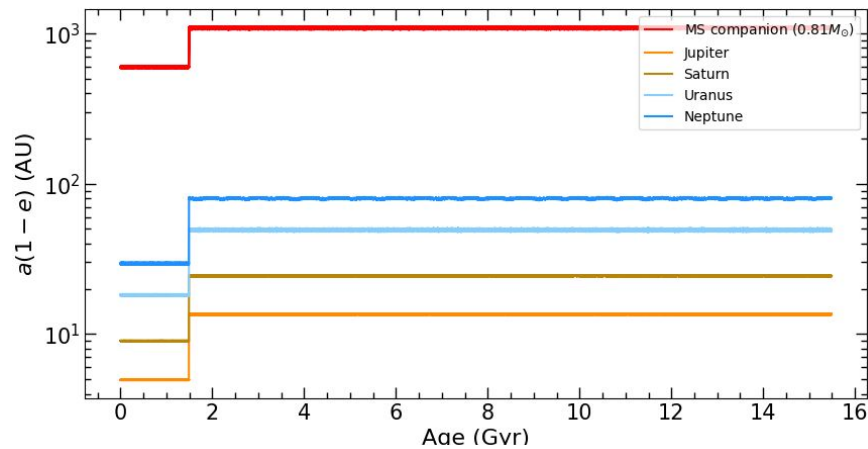
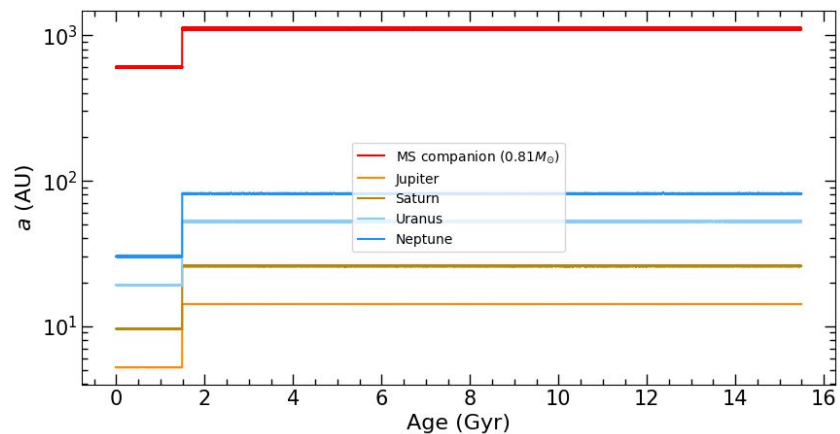
Results $2M_{\odot}$ no binary - WD $0.63M_{\odot}$



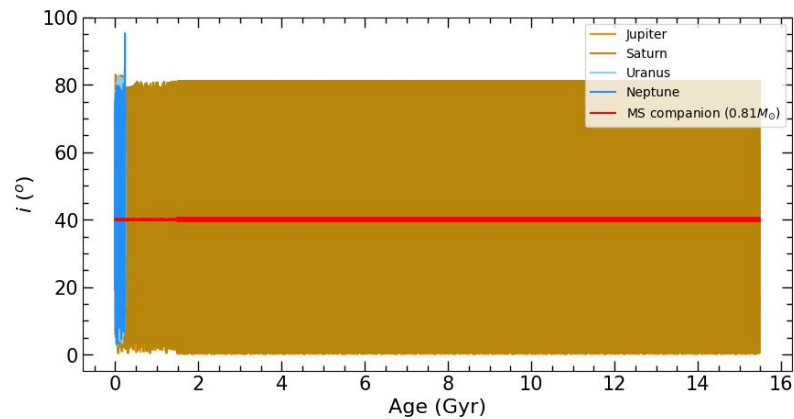
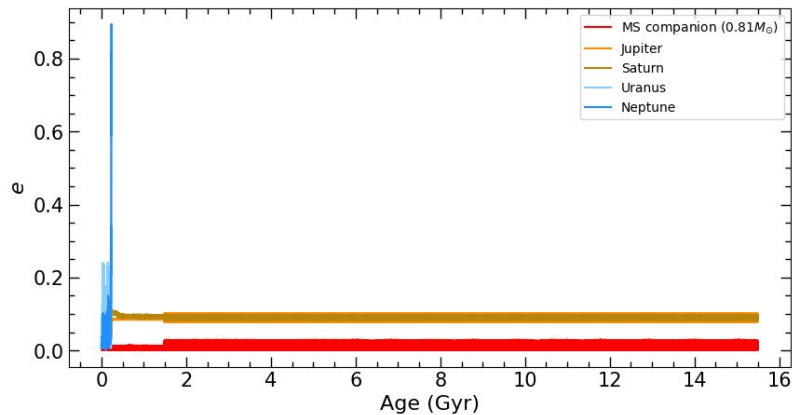
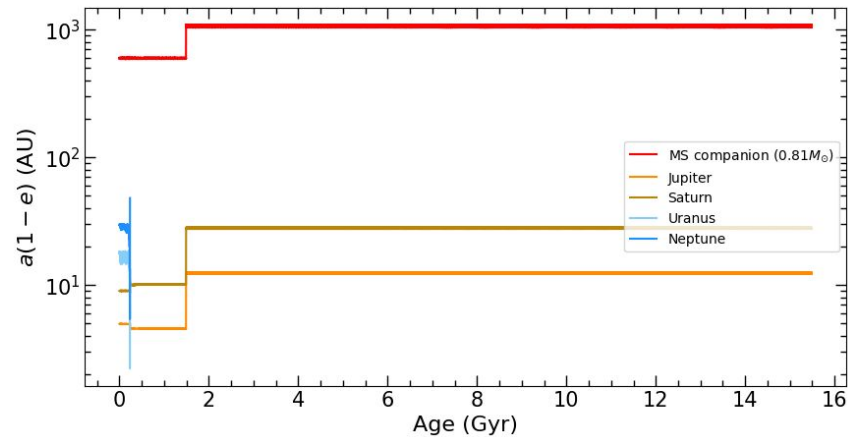
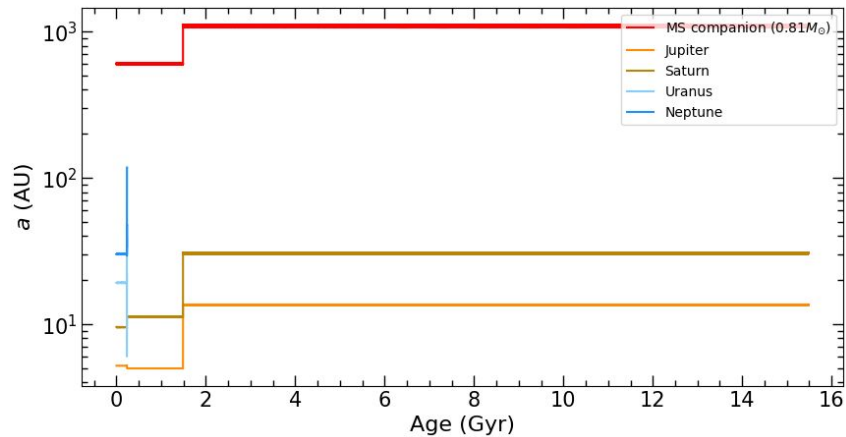
Results $2M_{\odot}$, $i=1^{\circ}$ - WD $0.63M_{\odot}$



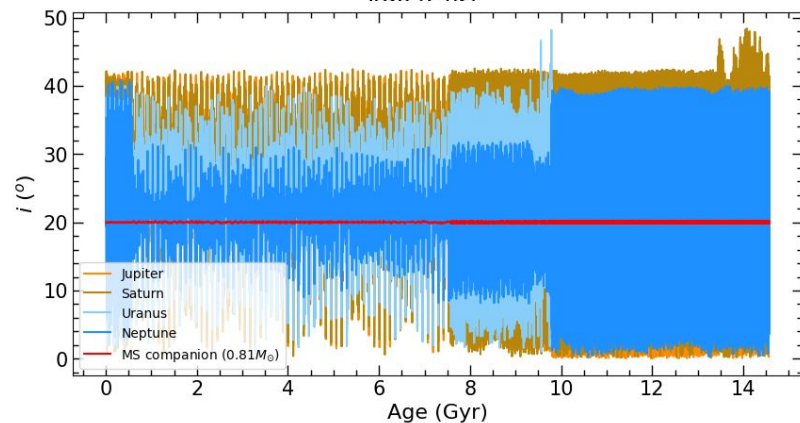
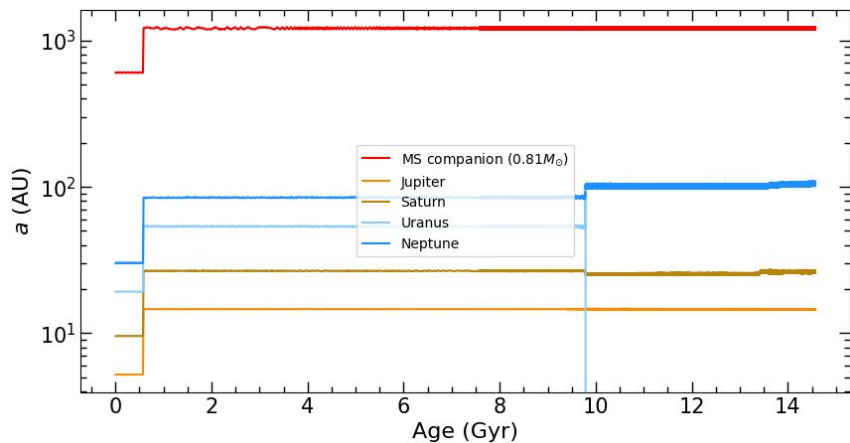
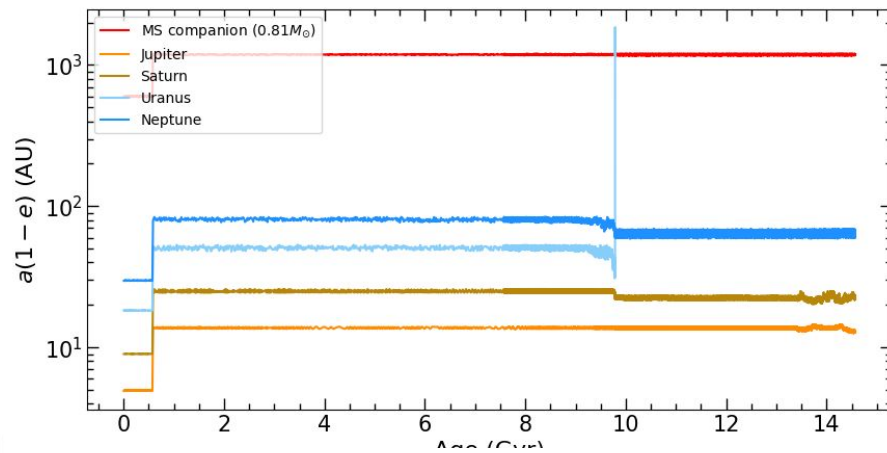
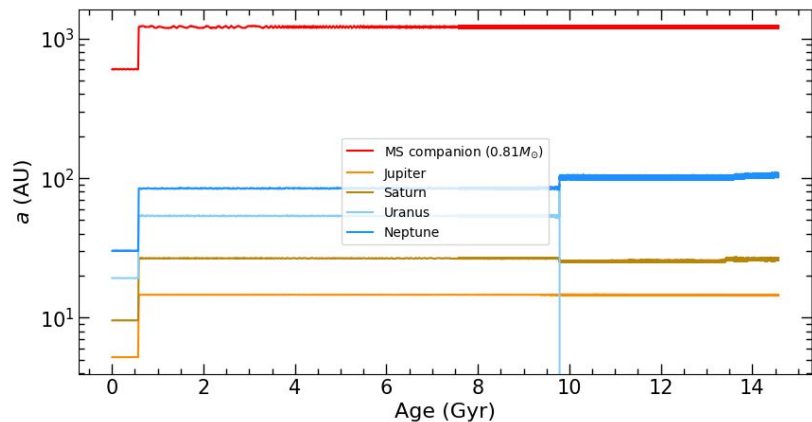
Results $2M_{\odot}$, $i=20^{\circ}$ - WD $0.63M_{\odot}$



Results $2M_{\odot}$, $i=40^{\circ}$ - WD $0.63M_{\odot}$



Results $3M_{\odot}$ $i=20^{\circ}$ - WD $0.72M_{\odot}$ (stable for no binary and $i=1^{\circ}$)



Results $3M_{\odot}$ $i=40^{\circ}$ - WD $0.72M_{\odot}$

